
Toscanini 05 transfer manifest

inv:fixturefluidics_hydraulic0e8d4aa7fcfb8 · run r541-ranked-pkg-battery-a · maturity
EARLY_TECHNICAL_EVALUATION · 3D class SYSTEM_3D

What is transferable

the recorded invention architecture for inv:fixture:fluidics_hydraulic:0:e8d4aa7fcfb8 with its evidence roles, the engineering projection (design inputs, failure modes, verification items, build plan, equations), the unknown roadmap, and the 3D artifacts this class earned (SYSTEM_3D)

Artifact inventory

every file with sha256 in PACKAGE_MANIFEST.json; the machine-readable layers listed in 00; MODEL/ carries the parametric source and derived CAD when ENGINEERING_3D is earned

Engineering maturity

EARLY_TECHNICAL_EVALUATION — basis and counts in MATURITY_BASIS.json

Buyer capability needed

an engineering team able to take ENGINEERING_PROPOSED state to manufacturable definition (the design work named in the unknown roadmap) and to run the pre-registered decisive experiment — the context-specific buyer capabilities are recorded in the buyer decision card (What must the buyer bring?), derived from the canonical applicability state

You must verify

the buyer carries the verification duty for every transferred claim (nothing arrives pre-verified):

- VF-001: bench shunt flow loop; measure flow decay over 30 days
- VF-002: run killer experiment: falsification_test_from_candidate (EIG/cost 0.44573539382492045)
- VF-003: bench flow-loop characterization across operating range
- VF-004: pressure-drop mapping vs flow (identify regime transitions)
- VF-005: delivered-flow trend below baseline in a flow-loop or animal model; pressure-head rise at constant flow (measured quantity: forward patency / free flow area)
- VF-006: kink/collapse test across duty cycle; flow-stoppage signature under applied load (measured quantity: mechanical stress / fatigue endurance)
- VF-007: pressure-flow mapping across the full envelope showing regime-break curvature (measured quantity: pressure / pressure-drop)
- VF-008: reverse-leakage rate measurement after accelerated cycling (measured quantity: reverse-flow / leakage rate)

Buyer and regulatory requirements (canonical applicability)

canonical problem context: MEDICAL_IN_VIVO

buyer type: medical device company (candidate buyer class for an in-vivo medical technology)
[APPLICABLE]

engineering capability: medical device design engineering with the domain's physics (see engineering specification governing model) [APPLICABLE]

manufacturing capability: medical-grade manufacturing (cleanroom assembly, medical extrusion/molding as the device requires) under a certified QMS [APPLICABLE]

manufacturing capability: sterilization-validation capability [APPLICABLE]

regulatory: medical-device regulatory pathway determination (FDA 510(k)/PMA or market jurisdiction equivalent) with biocompatibility evaluation (ISO 10993) and QMS (ISO 13485) [APPLICABLE]
market channels: clinical/hospital market channels [APPLICABLE]

Required tooling

CAD (STEP readers; CadQuery/OCCT for parametric rebuilds — see MODEL/PARAMETRIC_MODEL_SOURCE.py), mesh inspection (any STL viewer; trimesh used for the independent watertight check), and the bench equipment named in the build plan

Dependencies

no external services; the package is self-contained with real hashes (verify after transfer against PACKAGE_MANIFEST.json)

Confidentiality classification

CONFIDENTIAL — technical evaluation draft for the recipient's engineering diligence only; not a release document, not legal advice

Reproduction instructions

rebuild the geometry from MODEL/PARAMETRIC_MODEL_SOURCE.py + MODEL/PARAMETERS.json (the regeneration check records the expected measurements); verify every file against the manifest hashes

Open technical work

13 unknowns with resolution actions in UNKNOWN_ROADMAP.json — this is the recipient's first work queue

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.